Thesis Abstract: Aperture Synthesis as an Inverse Problem

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Radio interferometers measure samples of the Fourier transform of the sky brightness (visibilities) at the spatial frequencies set by the uv-coverage. Since this coverage is always incomplete, recovering an image from the visibilities is an ill-posed inverse problem, and any reconstruction must rely on prior information about the source. This work studies aperture synthesis within this framework and evaluates the speed and accuracy of four image synthesis methods, each built on a different prior, on simulated and archival ALMA data. The methods tested are CASA's `tclean` and WSClean, two implementations of the CLEAN algorithm; Basis Pursuit Sparsity Averaging (BPSA), which promotes sparsity in a dictionary of Daubechies wavelets; and R2D2, a series of deep neural networks that learns its prior from data. A new instance of R2D2, R2Disk2, was trained locally to image protoplanetary disks, on a dataset built with a disk simulator based on DSHARP morphologies: 4000 training and 600 validation images with their simulated ALMA Band 6 observations.

The methods were compared in three experiments. First, a hyperparameter sweep followed by evaluation on an independent test set showed that BPSA reaches the highest fidelity (PSNR of 46.7 dB), followed by R2Disk2 (37.7 dB), `tclean` (29.8 dB) and WSClean (29.3 dB). WSClean and R2Disk2 were the fastest (1.3 and 1.5 s per image), while BPSA was an order of magnitude slower (15.2 s). Second, varying the signal-to-noise ratio (SNR of 50, 200 and 500) revealed a ranking crossover: the CLEAN methods are the most robust (2–3 dB variation) and lead at the lowest SNR, while BPSA and R2Disk2 vary by 16–17 dB and dominate at high SNR (Figure 1). R2Disk2 degrades most at low SNR, outside its training regime. Third, the methods were applied to DSHARP observations of HD 163296 (Figure 2), Sz 114 and Sz 129. With its regularization tuned per target, BPSA recovered all the rings and gaps identified in the published images, including structure finer than the synthesized beam. R2Disk2 recovered the inner substructure in seconds, but removed the faint extended emission in the outer disks, a morphology absent from its training set. The CLEAN methods smoothed sub-beam features, with imaging times for `tclean` ranging from 48 s to 9 min per target and WSClean taking tens of seconds.

No method dominates in every regime: the best choice depends on the SNR of the data and the scientific goal. Learned methods such as R2D2 offer the best balance between fidelity and speed, making them promising for large surveys, provided their training covers the noise levels and morphologies of real observations.

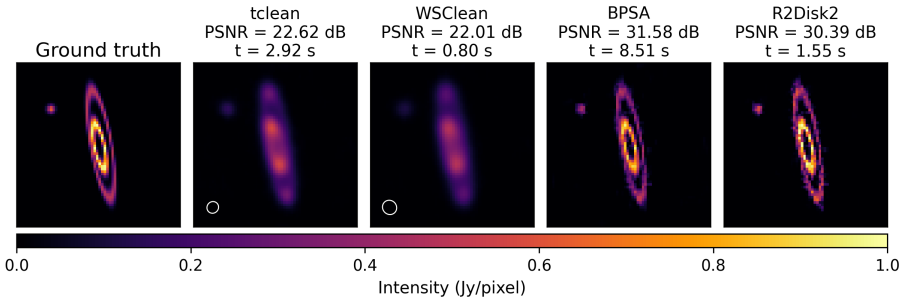


Figure 1: Illustrative example from the independent synthetic test set showing an inclined two-ring disk with an off-center point source at SNR = 200. The ground truth and all reconstructions are shown on a common intensity scale. PSNR values and imaging times correspond to this individual test case.

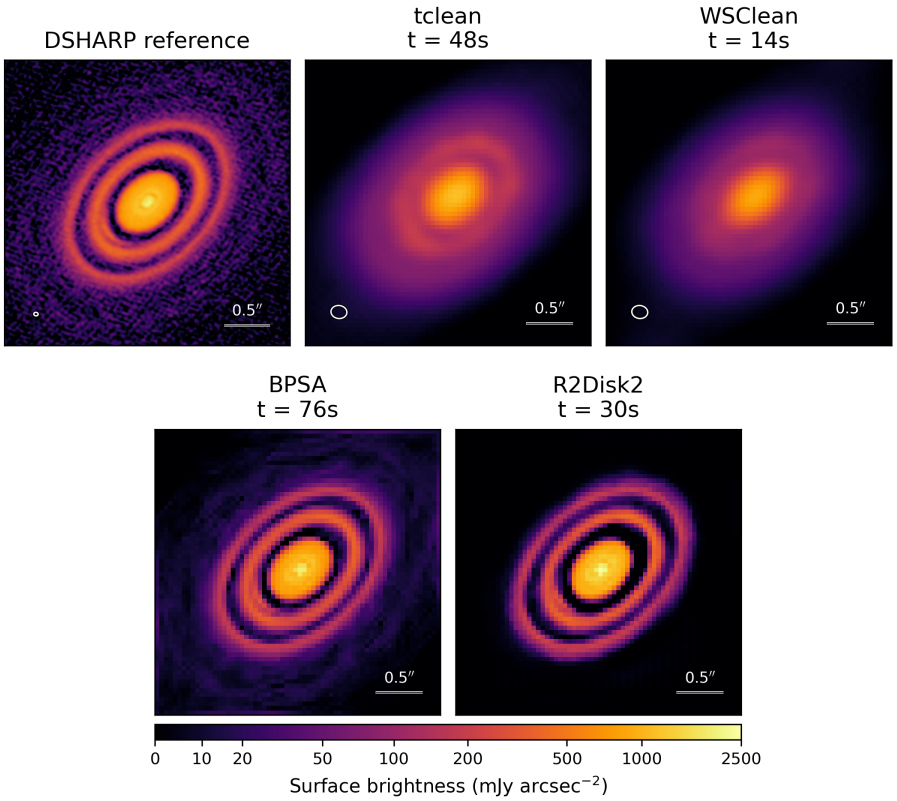


Figure 2: Reconstructions of archival ALMA Band 6 observations of HD 163296. The publicly available DSHARP fiducial FITS image (Andrews et al. 2018) is shown only as a visual reference. The reconstructions performed in this work use a 64 × 64 pixel grid. The CLEAN reconstructions used a basic setup with Briggs weighting and robust = 2, whereas the DSHARP image used different visibility weighting. The comparison is qualitative and not resolution matched. BPSA and R2Disk2 recover inner disk substructure, but R2Disk2 suppresses faint extended emission. All panels use a common asinh surface-brightness scale.